

EK369E: Financial Markets

Term Structure and Interest Rate Risk

Problem Set (with solutions)

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Problem 1

The table below summarises some information from the fixed-income market:

Bond	Maturity (years)	Coupon rate (%)	Yield (%)
A	5.5	0	2.5
B	2.5	4.0	2.0

- a) Calculate the duration and the modified duration for the two bonds.

Solution 1a

The duration of Bond A is equal to its time to maturity i.e. 5.5 yrs (because it's a zero-coupon bond)

The modified duration of Bond A: $D^* = \frac{5.5}{1.025} = \underline{5.37}$

Duration of Bond B: $\frac{255.2464}{106.8202} = \underline{2.39}$

Modified Duration of Bond B: $\frac{2.39}{1.02} = \underline{2.34}$

time	Cashflow	Discount factor at 2%	PV	$PV \times t$
0.5	4	1.0100	3.9606	1.9803
1.5	4	1.0301	3.8829	5.8244
2.5	104	1.0508	98.9767	247.4417
			106.8202	255.2464

- b) Which of the two bonds will have the smallest price change for a given change in the interest rate? Give a short explanation.

Solution 1b

For any 1% increase in interest rates:

Bond A will experience a 5.4% reduction in its price: $-5.37 \times 0.01 = -0.054$

Bond B will experience a 2.3% reduction in its price: $-2.34 \times 0.01 = -0.023$

Bond B has a shorter effective maturity (duration) and a higher coupon rate, therefore it is less vulnerable to future fluctuation in interest rates.

- c) If a fund manager invests (for face value) NOK 10 million in bond A and (for face value) NOK 16 million in bond B, what will the duration of the portfolio be?

Solution 1c

Price of Bond A: $\frac{100}{(1.025)^{5.5}} = 87.3$

Market Value of Bond A: NOK 10 million $\times \frac{87.3}{100} = \text{NOK } 8.73 \text{ million}$

Market Value of Bond B: NOK 16 million $\times \frac{106.82}{100} = \text{NOK } 17.09 \text{ million}$

Total Value of the Bond Portfolio: NOK 25.82 million

Weight of Bond A in the portfolio: $\frac{8.73\text{million}}{25.82\text{million}} = 0.34$

Weight of Bond B in the portfolio: $\frac{17.09\text{million}}{25.82\text{million}} = 0.66$

Duration of the portfolio: $(0.34 \times 5.5) + (0.66 \times 2.39) = \underline{\underline{3.44}}$

- d) Calculate the duration of the portfolio one year later, given that the yield is unchanged for both bonds.

Solution 1d

One year later

The duration of Bond A = 4.5yrs

Duration of Bond B: $\frac{153.4146}{104.9168} = \underline{\underline{1.46}}$

t	CF	DF	PV	$PV * t$
0.5	4	1.0100	3.96059	1.980295
1.5	104	1.0301	100.9562	151.4343
			104.9168	153.4146

e) What is duration drift, and how big is the duration drift for the two bonds?

Solution 1e

Duration drift refers to the fact that the change in the duration of a bond is less than the passage of time (or the change in the time to maturity). Furthermore, bonds with higher coupon rates will experience a faster decline in duration than a bond with a lower coupon rate. Zero-coupon bonds do not experience duration drift.

Bond	Initial duration	New duration	Δ Duration	Initial Maturity	New Maturity	Δ Maturity
A	5.5	4.5	1	5.5	4.5	1
B	2.39	1.46	0.93	2.5	1.5	1

Problem 2

In the fixed-income market, four different zero-coupon bonds are traded, all with a face value of NOK 100. The table below summarizes the relevant data:

Bond	Price	Years to Maturity
A	97.75	1
B	95.00	2
C	92.00	3
D	89.00	4

a) Calculate the rates on the spot yield curve i.e. s_1, s_2, s_3 and s_4 .

Solution 2a

$$\text{The 1-yr spot rate: } 97.75 = \frac{100}{1+r_1}$$

$$r_1 = \frac{100}{97.75} - 1 = \underline{\underline{2.30\%}}$$

$$\text{The 2-yr spot rate: } 95.00 = \frac{100}{(1+r_2)^2}$$

$$r_2 = \sqrt[2]{\frac{100}{95.0}} - 1 = \underline{\underline{2.60\%}}$$

The 3-yr spot rate: $92.00 = \frac{100}{(1+r_3)^3}$

$$r_3 = \sqrt[3]{\frac{100}{92.0}} - 1 = \underline{\underline{2.82\%}}$$

The 4-yr spot rate: $89.00 = \frac{100}{(1+r_4)^4}$

$$r_4 = \sqrt[4]{\frac{100}{89.0}} - 1 = \underline{\underline{2.96\%}}$$

c) A three-year coupon bond has a face value of NOK 100 and an annual coupon rate of 3%.

i Based on the yield structure above, what is the market price of the coupon bond?

ii Find the yield to maturity for the coupon bond.

iv Find the duration of the coupon bond.

Solution 2b

i.

Current market price of the 3% coupon bond with three years to maturity:

$$P_0 = \frac{3}{1.0230} + \frac{3}{1.0260^2} + \frac{103}{1.0282^3} = \underline{\underline{100.54}}$$

ii.

$$100.54 = \frac{3}{1+YTM} + \frac{3}{(1+YTM)^2} + \frac{103}{(1+YTM)^3}$$

$$YTM \approx \underline{\underline{2.81\%}}$$

iii.

$$\text{Duration of the bond: } \frac{292.944}{100.539} = \underline{\underline{2.91 \text{ yrs}}}$$

time	Cashflow	Discount factor	PV	$PV \times t$
1	3	1.0281	2.9180	2.9180
2	3	1.0570	2.8382	5.6765
3	103	1.0867	94.7832	284.3495
			100.5394	292.944

c) Calculate the implied one-year forward rates in year 2, year 3 and year 4.

Solution 2c

$$f_{1,2} = \frac{1.0260^2}{1.0230} - 1 = \underline{\underline{2.89\%}}$$

$$f_{2,3} = \frac{1.0282^3}{1.0260^2} - 1 = \underline{\underline{3.26\%}}$$

$$f_{3,4} = \frac{1.0296^4}{1.0282^3} - 1 = \underline{\underline{3.37\%}}$$

- d) Assuming that the spot yield curve evolves in line with the pure expectation hypothesis, compute the one-year holding period return on this bond.

Solution 2d

The yield curve one-year later:

$$\text{1-yr spot rate: } r_1 = f_{1,2} = 2.89\%$$

$$\text{2-yr spot rate: } (1 + r_2)^2 = (1 + r_1)(1 + f_{2,3})$$

$$r_2 = (\sqrt[2]{1.0289 \times 1.0326}) - 1 = \underline{\underline{3.08\%}}$$

$$\text{3-yr spot rate: } (1 + r_3)^3 = (1 + r_1)(1 + f_{2,3})(1 + f_{3,4})$$

$$r_3 = (\sqrt[3]{1.0289 \times 1.0326 \times 1.0337}) - 1 = \underline{\underline{3.18\%}}$$

The price of the 3% coupon bond one-year later:

$$P_1 = \frac{3}{1.0289} + \frac{103}{1.0308^2} = \underline{\underline{99.86}}$$

$$\text{Holding period return: HPR} = \frac{99.86 - 100.54 + 3}{100.54} = \underline{\underline{2.30\%}}$$

Problem 3

A government bond, Bond 1, with a face value of 100 and coupon rate of 4.50% will mature in four years. An investor is looking to invest NOK 5 million for a period of two years. The investor considers various strategies in the interest rate market. In addition to Bond 1, the investor may invest in government Bond 2. Bond 2s coupon rate is also 4.50%, but the time to maturity is two years. In the financial press, you see that the term structure (yield curve) for government bonds with a coupon rate of 4.50% is:

Year	1	2	3	4
Yield (%)	4.00	4.20	4.50	4.90

- a Assume that the yield curve is stable, calculate the geometric return for:

- i) Buy bond 1, and sell it after two years

ii) Buy bond 2, and hold it to maturity.

Solution 3a

i) Buy Bond 1 and sell it after two years

Price today @ t = 0:

$$P_0^1 = \frac{4.5}{1.049} + \frac{4.5}{1.049^2} + \frac{4.5}{1.049^3} + \frac{104.5}{1.049^4} = 98.58$$

Price one year later @ t = 1:

$$P_1^1 = \frac{4.5}{1.045} + \frac{4.5}{1.045^2} + \frac{104.5}{1.045^3} = 100$$

Price two years later @ t = 2:

$$P_2^1 = \frac{4.5}{1.042} + \frac{104.5}{1.042^2} = 100.56$$

Holding Period Return:

$$HPR_1^1 = \frac{100 - 98.58 + 4.5}{98.58} = 6\%$$

$$HPR_2^1 = \frac{100.56 - 100 + 4.5}{100} = 5.06\%$$

Geometric average return over investment period: $r_G^1 = \sqrt{1.06 \times 1.0506} - 1 = \underline{\underline{5.53\%}}$

ii) Buy bond 2, and hold it to maturity

Price today @ t = 0:

$$P_0^2 = \frac{4.5}{1.042} + \frac{104.5}{1.042^2} = 100.56$$

Price one year later @ t = 1:

$$P_1^2 = \frac{104.5}{1.040} = 100.48$$

$$\text{Holding Period Return: } HPR_1 = \frac{100.48 - 100.56 + 4.5}{100.56} = 4.39\%$$

Price two years later @ t = 2:

$$P_2^2 = 100 \text{ (bond is redeemed at par value)}$$

$$\text{Holding Period Return: } HPR_1^2 = \frac{100.48 - 100.56 + 4.5}{100.56} = 4.39\%$$

$$HPR_2^2 = \frac{100 - 100.48 + 4.5}{100.48} = 4.00\%$$

Geometric average return over investment period: $r_G^2 = \sqrt{1.0439 \times 1.04} - 1 = \underline{\underline{4.2\%}}$

b) Explain why there are differences between the two strategies in a)

Solution 3b

Investment strategy 1 is called rolling down or riding the yield curve while Investment strategy 2 is known as Maturity Matching. When the yield curve is upward-sloping and stable, investment strategy 1 offers superior returns over strategy 2.

Problem 4

In the fixed-income market, we have the following information for zero-coupon bonds:

Maturity (years)	1	2	3
Yield to maturity (%)	4.3	4.0	3.8

You are analysing three bonds, all maturing in three years, but with different coupon rates. Bond A has a coupon rate of 8%, bond B has a coupon rate of 4%, while bond C is a zero coupon bond.

a) Calculate the price of the three bonds.

Solution 4a

Price of Bond A:

$$P_0^A = \frac{8}{1.043} + \frac{8}{1.04^2} + \frac{108}{1.038^3} = \underline{\underline{111.63}}$$

Price of Bond B:

$$P_0^B = \frac{4}{1.043} + \frac{4}{1.04^2} + \frac{104}{1.038^3} = \underline{\underline{100.52}}$$

Price of Bond C:

$$P_0^C = \frac{100}{1.038^3} = \underline{\underline{89.41}}$$

b) Without making calculations, briefly explain which bond has the lowest yield to maturity and which bond has the lowest duration.

Solution 4b

Holding time to maturity constant: Bond C offers the lowest average return at maturity compared to other bonds. This is because Bond C is a zero-coupon bond with a single cash flow offered at maturity. In contrast, Bond A offers the highest average return to the bond investors at maturity. Bond A has the lowest effective duration. With the highest coupon rate, Bond A offers more of its total cash flow at earlier periods in the bond's life. In contrast, Bond C has the highest duration since all of its cash flow is offered at maturity.

You have an investment horizon of two years and are considering investing in either bond A or bond C. You are considering two different strategies: **i)** Buy bond A today, reinvest the coupon payment in one year in the same security and sell in two years. **ii)** Buy bond C today and sell it in two years. You assume that the yield curve given in the table above is stable.

c) Calculate the annual geometric average return for each of the two investment strategies.

Solution 4c

i) Buy bond A today, reinvest the coupon payment in one year in the same security, and sell in two years

$$P_0^A = \underline{111.63}$$

$$P_1^A = \frac{8}{1.043} + \frac{108}{1.040^2} = \underline{107.52}$$

$$P_2^A = \frac{108}{1.043} = \underline{103.55}$$

Description	Cash flow
1st coupon received at t = 1 and reinvested	$8 \times 1.04 = 8.32$
2nd coupon received at t = 2	8
Proceeds from bond sale	103.55
Settlement amount	119.87

Invested amount (V_0) is NOK 111.63

Settlement amount (V_2) is NOK 119.87

$$\text{Geometric/Time Weighted/ Compound Return} = \sqrt[2]{\frac{119.87}{111.63}} - 1 = \underline{3.62\%}$$

ii) Buy bond C today and sell it in two years

Price of Bond B:

$$P_0^C = \frac{100}{1.038^3} = \underline{89.41}$$

$$P_1^C = \frac{100}{1.040^2} = \underline{\underline{92.46}}$$

$$P_2^C = \frac{100}{1.043} = \underline{\underline{95.88}}$$

$$\text{Holding Period Return (HPR): } HPR_1 = \frac{92.46 - 89.41}{89.41} = 3.4\%$$

$$HPR_2 = \frac{95.88 - 92.46}{92.46} = 3.7\%$$

$$\text{Geometric return: } \sqrt{1.034 \times 1.037} - 1 = \underline{\underline{3.55\%}}$$

d) Which alternative is best? Briefly explain why.

Solution 4d

Strategy 2 (Maturity Matching) offers better returns than Strategy 1 (Rolling down the yield curve). Maturity matching ensures that capital gains offset low reinvestment returns when interest rates fall and vice versa. Since the yield curve is downward sloping, which indicates that interest rates are falling, Strategy 2 offers a better return to the investor. If the yield curve was upward-sloping, strategy 1 would perform better than Strategy 2.

e) Is there an alternative investment strategy in this market that would be clearly better than the two mentioned here?

Solution 4e

With a downward-sloping yield curve, bonds with longer maturity sell at a higher price than bonds with short maturity. Also, zero coupon bonds are attractive when interest rates are low since the investor enjoys capital appreciation without reinvestment risk. In this case, the best strategy would be to adopt a roll-over strategy i.e. buying two consecutive 1-yr zero coupon bonds. The geometric return from such a strategy would be $(\sqrt[2]{1.043 \times 1.04}) - 1 = 4.15\%$, which outperforms Strategy 1 and Strategy 2.

COMMENTS:

If Bond D is the 1-yr zero coupon bond available at $t = 0$ and Bond E is the 1-yr zero coupon bond

available at $t = 1$

$$P_0^D = \frac{100}{1.043} = \underline{95.88}$$

$$P_1^D = 100 \text{ (Redemption of bond at par value)}$$

$$\text{Holding period return from investing in Bond D: } HPR_1 = \frac{100 - 95.88}{95.88} = 4.30\%$$

$$P_1^E = \frac{100}{1.040} = \underline{96.15}$$

$$P_2^E = 100 \text{ (Redemption of bond at par value)}$$

$$\text{Holding period return from investing in Bond E: } HPR_2 = \frac{100 - 96.15}{96.15} = 4.00\%$$

You put together a portfolio consisting of a face value of NOK 100 million in bond A and a face value of NOK 91.9 million in a one-year zero-coupon bond D.

f) What is the duration of this portfolio?

Solution 4f

Duration of Bond D = its time to maturity = 1 yr

$$\text{Duration of Bond A: } \frac{312.1661}{111.634} = \underline{2.796 \text{ yrs}}$$

time	Cashflow	Discount factor = $[(1 + r)^t]$	PV	$PV \times t$
1	8	1.043	7.6702	7.6702
2	8	1.0816	7.3965	14.7929
3	108	1.1184	96.5677	289.703
			111.634	312.1661

$$\text{Purchase amount for Bond A: NOK 100 million} \times \frac{111.634}{100} = \text{NOK 111.634 million}$$

$$\text{Purchase amount for Bond D: NOK 91.9 million} \times \frac{95.88}{100} = \text{NOK 88.114 million}$$

$$\text{Value of Bond Portfolio: NOK 111.634 million} + \text{NOK 88.114 million} = \text{NOK 199.748 million}$$

$$\text{Weight of Bond A in the portfolio: } \frac{111.634}{199.748} = 0.56$$

$$\text{Weight of Bond B in the portfolio: } \frac{88.114}{199.748} = 0.44$$

$$\text{Duration of the bond portfolio: } (0.56 \times 2.796) + (0.44 \times 1) = \underline{2.00 \text{ yrs}}$$

Problem 5

A newly issued bond has a maturity of 10 years and pays a 7% coupon rate annually. The bond sells at par value.

a) What is the duration of the bond?

Solution 5a

t	CF	DF at 7%	PV	$PV \cdot t$
1	7	1.07	6.5421	6.5421
2	7	1.1449	6.1141	12.2281
3	7	1.2250	5.7141	17.1423
4	7	1.3108	5.3403	21.3611
5	7	1.4026	4.9909	24.9545
6	7	1.5007	4.6644	27.9864
7	7	1.6058	4.3592	30.5147
8	7	1.7182	4.0741	32.5925
9	7	1.8385	3.8075	34.2678
10	107	1.9672	54.3934	543.9337
			100	751.5232

Bond Duration: $\frac{751.5232}{100} = \underline{\underline{7.52 \text{ yrs}}}$

or

t	CF	DF at 7%	PV	$w = \frac{PV_t}{P_0}$	$w \cdot t$
1	7	1.07	6.542056	0.065421	0.065421
2	7	1.1449	6.114071	0.061141	0.122281
3	7	1.225043	5.714085	0.057141	0.171423
4	7	1.310796	5.340266	0.053403	0.213611
5	7	1.402552	4.990903	0.049909	0.249545
6	7	1.50073	4.664396	0.046644	0.279864
7	7	1.605781	4.359248	0.043592	0.305147
8	7	1.718186	4.074064	0.040741	0.325925
9	7	1.838459	3.807536	0.038075	0.342678
10	107	1.967151	54.39337	0.543934	5.439337
			100	7.515232	

b) What is the convexity of the bond?

Solution 5b

t	CF	DF	PV	$t^2 + t$	$PV \cdot (t^2 + t)$
1	7	1.07	6.5421	2	13.08411
2	7	1.1449	6.1141	6	36.68443
3	7	1.2250	5.7141	12	68.56902
4	7	1.3108	5.3403	20	106.8053
5	7	1.4026	4.9909	30	149.7271
6	7	1.5007	4.6644	42	195.9046
7	7	1.6058	4.3592	56	244.1179
8	7	1.7182	4.0741	72	293.3326
9	7	1.8385	3.8075	90	342.6783
10	107	1.9672	54.3934	110	5983.271
					7434.175

Convexity of the bond: $\frac{1}{100} \cdot \frac{1}{1.07^2} \cdot 7434.175 = \underline{\underline{64.93}}$

If the interest rate in the market is expected to increase from 7% to 8%;

c) What new price is predicted by the modified duration rule

Solution 5c

Modified duration: $\frac{7.52}{1.07} = 7.02$

price risk: $-7.02 \cdot 1\% = -7.02\%$

New price predicted by the bond's duration: $\text{NOK } 100 \times 0.9298 = \underline{\underline{\text{NOK } 92.98}}$

If interest rates are expected to increase from 7% to 8%

d) What new price is predicted by the convexity rule

Solution 5d

Price risk: $(-7.02 \times 0.01) + (0.5 \times 64.93 \times 0.01^2) = -0.0669$ or -6.7%

New price predicted by the duration and convexity of the bond: $\text{NOK } 100 \times 0.9330 = \underline{\underline{\text{NOK } 93.30}}$

e) Find the actual price assuming that interest rates immediately increase from 7% to 8% (time to maturity is still 10 years)

Solution 5e

$$P_0 = 7 \left[\frac{1 - (1.08^{-10})}{0.08} \right] + \frac{100}{(1.08)^{10}} = \underline{\underline{93.29}}$$

Problem 6

A 6% coupon bond paying interest annually has a modified duration of 10-yrs, sells for NOK 80 and is priced at a yield to maturity of 8%. If the YTM increases to 9%, what is the predicted change in price based on the bond's duration

Solution 6

Price risk: $-10 \cdot 1\% = -10\%$

New price predicted by modified duration: $\text{NOK } 80 \times 0.90 = \underline{\underline{\text{NOK } 72}}$

Problem 7

A pension plan has an obligation to pay NOK 10,000 once a year during a 10-year period, beginning 5 years from now. The interest rates are flat at 10%

a) what is the duration of this obligation?

Solution 7a

The duration of the obligation is 9.60 yrs.

t	CF	DF	PV	w	w*t
5	10000	1.61051	6209.21	0.162746	0.813728
6	10000	1.771561	5644.74	0.14795	0.887703
7	10000	1.948717	5131.58	0.1345	0.941503
8	10000	2.143589	4665.07	0.122273	0.978185
9	10000	2.357948	4240.98	0.111157	1.000416
10	10000	2.593742	3855.43	0.101052	1.010522
11	10000	2.853117	3504.94	0.091866	1.010522
12	10000	3.138428	3186.31	0.083514	1.00217
13	10000	3.452271	2896.64	0.075922	0.986986
14	10000	3.797498	2633.31	0.06902	0.96628
				41,968.2	9.598013

b) If the pension fund intends to use a 5-yr and 20-yr zero coupon bonds to immunise this position, how much should be placed in each bond?

Solution 7b

Duration of a 5-yr zero coupon bond = 5 yrs

Duration of a 20-yr zero coupon bond = 20 yrs

Amount invested in the 5-yr zero coupon bond: w

Amount invested in the 20-yr zero coupon bond: $1 - w$

$$9.6 = 5w + 20(1 - w)$$

$$w = 0.69$$

$$1 - w = 0.31$$

Market value investment in each bond:

$$\text{5-yr bond: } 0.69 \times 41968.2 = \text{NOK } 29,098$$

$$\text{20-yr bond: } 0.31 \times 41968.2 = \text{NOK } 12,871$$

Face value investment in each bond:

$$\text{5-yr bond: } \text{NOK } 29,098 \times 1.1^5 \approx \text{NOK } 46863$$

$$\text{20-yr bond: } \text{NOK } 12871 \times 1.1^{20} \approx \text{NOK } 86,585$$

Problem 8

The 6-month Treasury bill spot rate is 4% and the 1-yr treasury bill spot rate is 5%. What is the implied 6-month forward rate for six months from now?

Solution 8

$$1 + (f_{0.6,1} \times 0.6) = \frac{1 + (0.05)}{1 + (0.04 \times 0.6)}$$

$$f_{0.6,1} = \underline{\underline{4.23\%}}$$

Problem 9

Below is a list of prices for zero coupon bonds with par values of NOK 100.

Maturity (Yrs)	Price (%)
1	94.34
2	87.35
3	81.64

- a) An 8.5% coupon bond making annual coupon will mature in three years. What should the yield to maturity on the bond be?

- b) if at the end of the first year, the yield curve flattens out at 8%. What will be the 1-yr holding period return on the coupon bond?

Solution 9a

The price of the the 8.5% coupon bond is NOK 104.02

The yield to maturity of the 8.5% coupon bond is 6.97%

- b) if at the end of the first year, the yield curve flattens out at 8%. What will be the 1-yr holding period return on the coupon bond?

Solution 9b

$$P_0 = 104.02$$

$$P_1 = 100.89$$

$$\text{HPR} = 5.16\%$$

Problem 10

The current yield curve for default-free zero-coupon bonds is as follows.

Maturity (Yrs)	Yield to Maturity (%)
1	10.00
2	11.00
3	12.00

Assuming that the pure expectation hypothesis of the term structure is correct and market expectations are accurate, what will be the yield to maturity on 1-year zero coupon bonds next year?

Solution 10

Under the pure expectation hypothesis, the yield to maturity of 1-year zero coupon bonds next year (s_2) is equal to the one-year implied forward rate in year 2 ($f_{1,2}$) = 12%